

AJAY VENKATESH

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Education

New York University, New York, NY

Aug 2024 – May 2026

Master of Science in Computer Engineering (GPA: 3.9/4.0)

Coursework: System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Experience

• Campus Mesh

Jan 2026 – Present

Software Engineer

New York, US

- Developed scalable **Node.js microservices** powering real-time academic workflows for **10K+ users** across study groups, messaging, notifications, and profile management, backed by a **MongoDB** store.
- Built a personalized **recommendation engine** using collaborative filtering and behavioral analytics, increasing engagement by **28%** and retention by **50%**.
- Worked on **LLM-based retrieval pipelines (RAG)** for CampusIQ (AI Assistant), leveraging **embedding-based semantic search** and structured data indexing to improve contextual accuracy by **30%**.

• Amazon

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, US

- Deployed a **Retrieval-Augmented Generation (RAG)** pipeline with an **agentic workflow** that autonomously retrieves operational logs to reason over incident details, generating context-aware summaries and remediation steps.
- Implemented a **Model Context Protocol (MCP)** server for efficient tool-augmented retrieval, cutting AWS Athena query time and compute costs by **40%**.
- Built a high-throughput, event-driven incident triage system on **AWS ECS Fargate, SQS, SNS**, decoupling distributed message processing and reducing response latency by **50%**.
- Provisioned resilient **Infrastructure-as-Code** using **AWS CDK** on auto-scaling ECS clusters, ensuring **99.9%** availability under 2x peak load.

• New York University (Novel AI Technologies, Inc.)

Aug 2024 – Present

Software Developer

New York, US

- Architected an **NSF-funded (\$100K)** AI-powered health app for **iOS** and **Android** processing real-time wearable data; deployed cloud-hosted AI models for health metric analysis and food image processing.
- Engineered **ETL pipelines** transforming **NoSQL** into **PostgreSQL** with **concurrency control and locking**; processed **250K+** records and powered **AWS QuickSight** dashboards.
- Developed an insurance analytics platform (**React, Node.js, REST APIs**) deployed via **Docker** on **EC2**; implemented Stripe billing, RBAC, and row-level security, generating **\$50K+** revenue across **6 clients**.
- Streamlined underwriting via a broker portal integrated with insurer platforms; implemented **document/media processing** and **WebSocket-based real-time communication**.

• PricewaterhouseCoopers (PwC)

Aug 2021 – Aug 2024

Senior Software Engineer

Bengaluru, India

- Developed a distributed backend on **Microsoft Azure** for a health platform integrating with **Microsoft CRM**, with event-driven processing and fault-tolerant retries over **SQL Server** stores.
- Deployed an **NLP-based OCR neural model** on Azure to automate structured data extraction, reducing processing time by **75%**.
- Optimized data ingestion with **parallel processing** on **100K-row** Excel feeds, cutting execution from **5 hours to 1 hour**.
- Led code reviews and mentored engineers on **software design principles**, reducing production bugs by **25%**.

Projects

• InviLite | AWS, SageMaker, Bedrock, FAISS, Python

- Architected a cloud-native data pipeline (Kinesis, Lambda, S3, Glue, DynamoDB) ingesting structured and unstructured enterprise data into governed ML pipelines.
- Built a **RAG system** (SVM + FAISS + **Bedrock LLM**), cutting incident resolution from **1–2 hours to minutes** and lifting root-cause classification accuracy from **56% to 92%**.

• RoBERTa Fine-Tuning with Custom LoRA | Python, PyTorch, Hugging Face, AGNews

- Architected a **parameter-efficient fine-tuning (PEFT)** pipeline using a custom **Low-Rank Adaptation (LoRA)** implementation to optimize a **RoBERTa-base** model for news classification.
- Optimized model efficiency by freezing pretrained weights and injecting trainable rank-decomposition matrices, reducing trainable parameters to under **1M (934K)** while achieving **83.45%** accuracy on the private leaderboard.
- Conducted extensive hyperparameter sweeps for rank (r) and alpha (α) values, utilizing **AdamW** and linear decay schedulers to stabilize training under strict memory and parameter constraints.

• High-Efficiency ResNet Image Classifier | Python, PyTorch, Torchvision, Matplotlib

- Designed and implemented a specialized **ResNet** architecture optimized for the **CIFAR-10** dataset, adhering to a strict **5M parameter** limit for edge-ready deployment.
- Enhanced model convergence and generalization by integrating the **Lookahead optimizer** and advanced data augmentation techniques, including horizontal flips and random cropping.
- Developed a modular training and evaluation framework with real-time visualization via **TensorBoard**, managing checkpoints and learning rate scheduling for reproducible experimentation.

Skills

- **Programming Languages:** C++, Java, Python, TypeScript, JavaScript (ES6), SQL
- **Web & Backend:** React, Node.js, Express.js, REST API, GraphQL API, Microservices, HTML, CSS, Sequelize
- **Databases:** MySQL, PostgreSQL, DynamoDB, MongoDB, Firebase, Dataverse
- **Cloud & DevOps:** AWS (CDK, EC2, S3, DynamoDB, API Gateway, Lambda, Cognito, IAM, SageMaker), Azure (OpenAI, Function Apps, Storage), Docker, Kubernetes, CI/CD
- **AI/ML:** PyTorch, JAX, Dynamo, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning, Reinforcement Learning, Hugging Face, RAG systems
- **Tools & Platforms:** Git, Linux, Visual Studio, Cursor, GitHub Copilot, Jupyter, Postman, Power BI

Publications

Supervised ML System Based Segmentation and Classification of Stroke Using Deep Learning Techniques (*IEEE*)

An Aquila-optimized SVM Classifier for Diabetes Prediction (*IEEE*)

Certifications

AWS Cloud Practitioner

Stanford Machine Learning (Coursera)

Azure Data Fundamentals